

Let $x_3 \neq 0$. $\therefore a_3 = -(x_1/x_3)a_1 + (-x_2/x_3)a_2$.

This is a contradiction to the fact that $a_3 \notin \text{Span}\{a_1, a_2\}$

$$\therefore x_3 = 0. \quad x_1 a_1 + x_2 a_2 + 0 \cdot a_3 = 0 \Rightarrow x_1 a_1 + x_2 a_2 = 0. \quad \therefore x_1 = x_2 = 0.$$

So, the matrix equation $Ax=0$, where $A=[a_1 a_2 a_3]$ has the unique trivial solution only.

By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

By the row reduction algorithm and Theorem 1, the given augmented matrix $[A \ 0]$ can be converted to a unique ref form.

By Theorem 2, the matrix A has no free variables. By defn, each of the 3 columns a_1, a_2 and a_3 have pivots.

Let, Row 1 Column 1 have a pivot. $\therefore$ Row 1 Column 1 = Π

From Cond. 3 of the matrix to be in echelon form, Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0.

Suppose Row 2 has no pivots. $\therefore$ Row 2 Col 2 = Row 2 Col 3 = 0

Since Col 2 and Col 3 must have a pivot, it should be present in Row 3 and Row 4. This violates condn 1 of the matrix to be in echelon form.

$\therefore$ Row 2 must have a pivot.

Let, Row 2 Column 3 have a pivot. $\therefore$ Row 2 Column 3 = Π

Col 2 must have a pivot in Row 3 or Row 4. This violates condn. 2 of the matrix to be in echelon form.

$\therefore$ Row 2 Column 3 has no pivot. $\therefore$ So, Row 2 Column 2 must have a pivot.

$\therefore$ Row 2 Column 2 = Π .

By Cond. 3 of the matrix to be in echelon form, Row 3 Column 2 = Row 4 Column 2 = 0.

There are no constraints on the positions of Row 1 Col 2, Row 1 Col 3, ~~Row 1 Col 4~~

~~Row 2 Col 3, Row 2 Col 4~~

$\therefore$ Row 1 Column 2 = Row 1 Column 3 = ~~Row 1 Column 4~~ = Row 2 Column 3 = ~~Row 2 Column 4~~

~~Column 4~~ = *